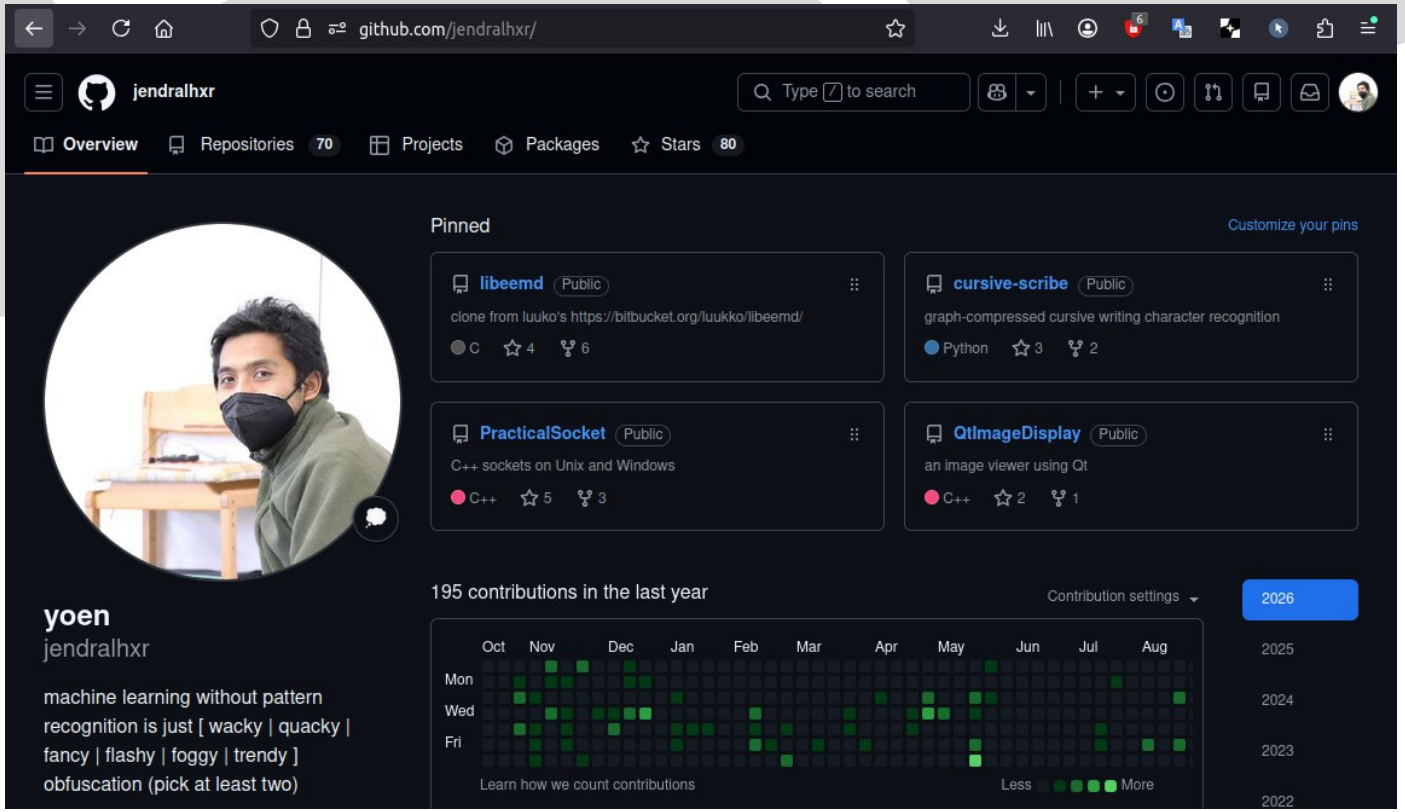
Introduction to Data Science

making sense from within the mess, one plot at a time



- B.Eng. (Computer Engineering, Department of Electrical Enginering, Brawijaya University, Indonesia) *parallel computing of higher-order integrals using homebrew computer cluster*
- M.Eng. (Department of System Cybernetics, Hiroshima University, Japan): *image reconstruction of projected imagery*
- Dr.Eng. (Department of System Cybernetics, Hiroshima University, Japan): *camera-based vibration displacement measurement for bridge structural health monitoring*
- PD (Research Center of Artificial Intelligence and Cybersecurity, National Research and Innovation Agency, Indonesia): *digital curation and character recognition for ancient Jawi manuscripts*
- Lecturer (Department of Data Science, Universitas Pembangunan Nasional Veteran Jawa Timur, Indonesia): *physical measurement-based data analysis and digital image processing*

Definition

- Data science combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI) and machine learning with specific subject matter expertise to uncover actionable insights hidden in an organization’s data (IBM)
- A field of study that uses scientific methods, processes, and systems to extract knowledge and insights from data, intersection between statistics and computer science (Harvard)

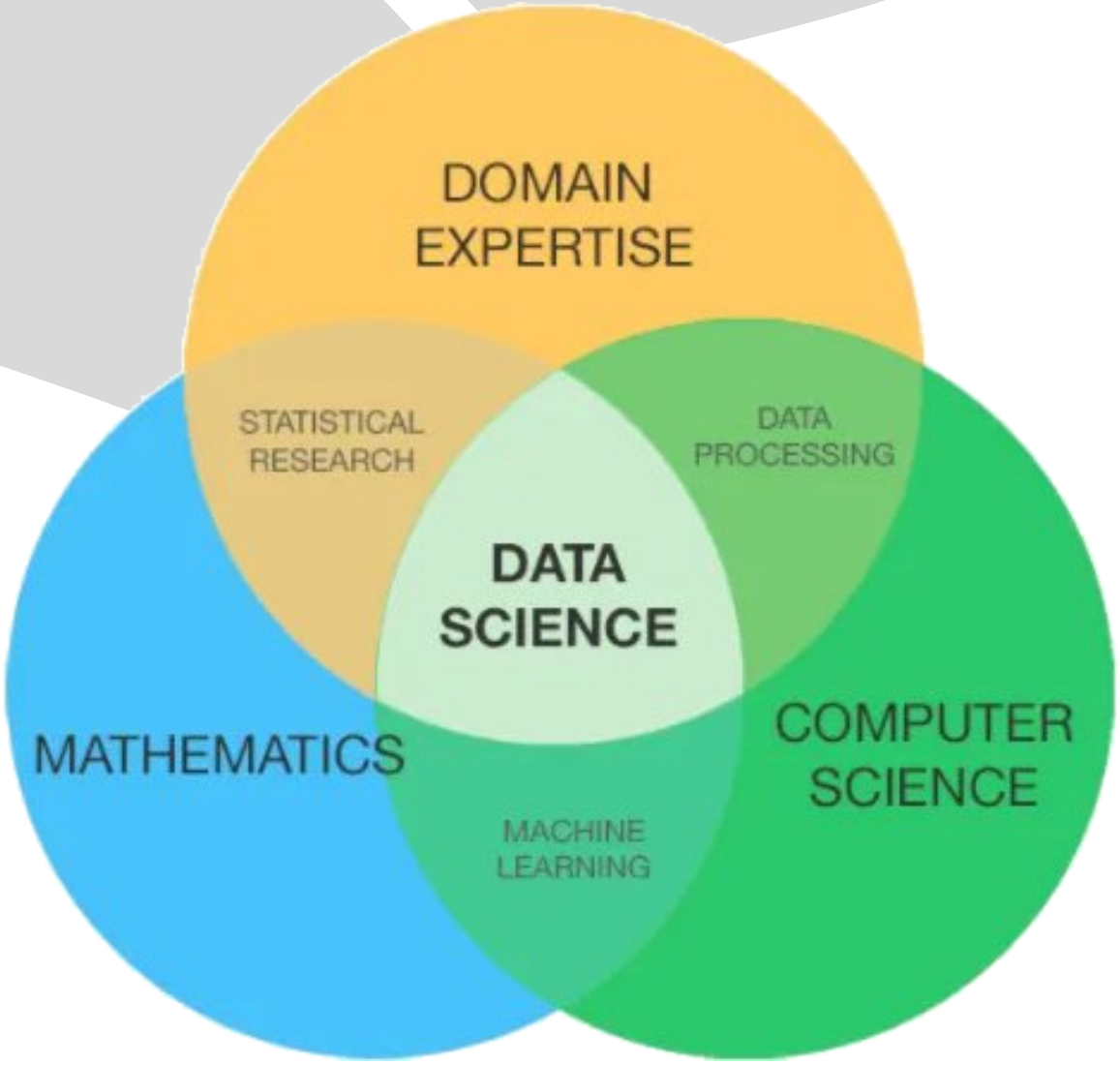


Illustration on the Harvard's

https://medium.com/@ahmedabubakr92/part-1-data-science-overview-41de2702fa09

Brief History

- 1962: John Tukey: data *analysis* (the Father of FFT)
- 1973: Peter Naur: *datalogy*, data *science* (ALGOL)
- ...
- 1998: Google
- 2009: Wolfram Alpha
- 2015: OpenAI, ChatGPT
- 2023: Claude

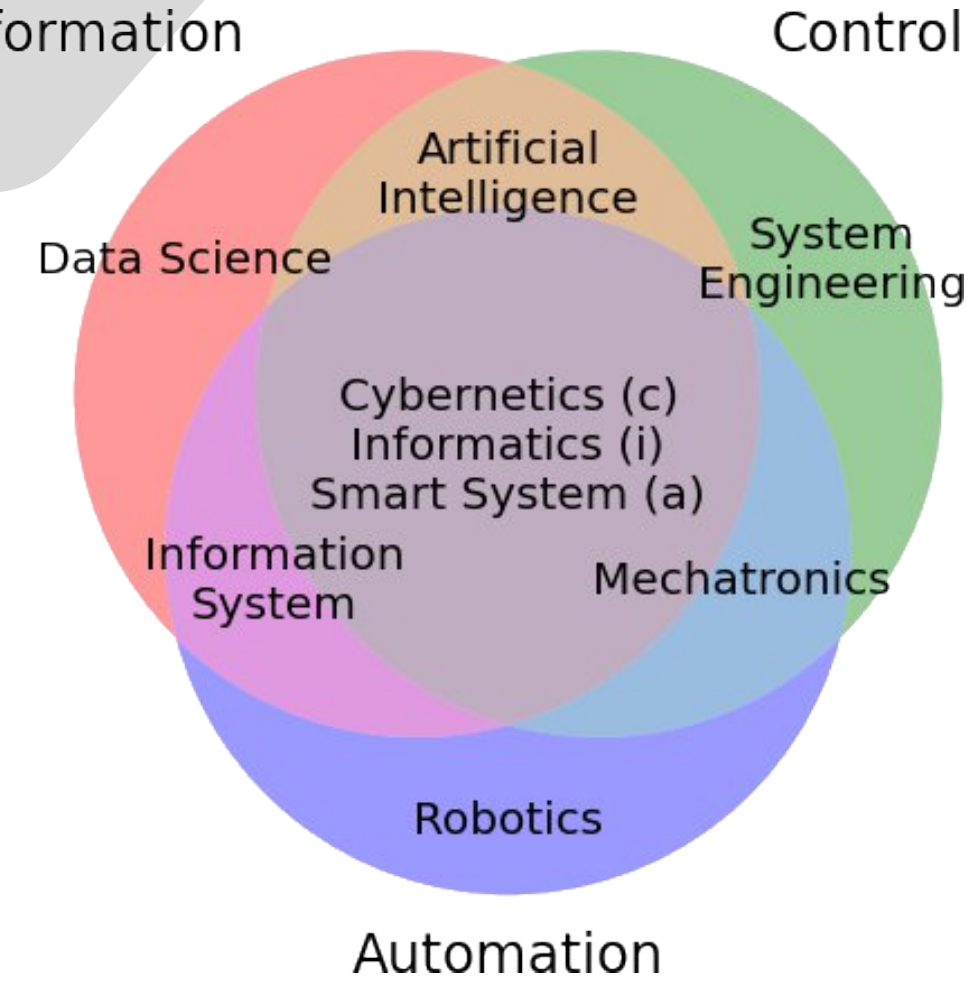
In relation to other disciplines, theoretical PoV

Aspect	Data Science	Data Analytics	Business Intelligence (BI)
Primary Focus	Predicting the future, finding hidden patterns, and building AI/automation models.	Analyzing historical data to answer specific business questions and identify trends.	Monitoring current and past business performance using reports and visual dashboards.
Questions Answered	*"What will happen?"* or *"How can we automate this?"*	*"Why did this happen?"* or *"What are the underlying trends?"*	*"What is happening right now?"* or *"What were our sales last month?"*
Time Horizon	Future-oriented (Predictive & Prescriptive).	Past & Present (Diagnostic & Descriptive).	Past & Present (Descriptive & Historical).
Data Types	Structured, Unstructured, and Semi-structured (Text, Images, Audio, Big Data).	Mostly Structured & Semi-structured Data.	Mostly Structured Data (Relational Databases, Data Warehouses).
Methods & Techniques	Machine Learning, Deep Learning, Statistical Modeling, Natural Language Processing (NLP).	Statistical Analysis, Hypothesis Testing, Diagnostic Analytics, Data Wrangling.	Data Aggregation, KPI Tracking, Reporting, Data Visualization, OLAP.
Primary Tools	Python, R, TensorFlow, PyTorch, Spark, Jupyter Notebooks.	Python, R, SQL, Excel	Power BI, Tableau, Looker, Excel, SQL, Qlik.
Core Skillset	Advanced programming, high-level math & statistics, ML algorithms.	Statistical analysis, SQL querying, data cleaning, basic visualization.	Business data modeling, dashboarding, KPI strategy, reporting.
Key Deliverable (Output)	Machine Learning models, recommendation engines, predictive APIs.	In-depth analytical reports, actionable insights, strategic recommendations.	Interactive dashboards, executive reports, KPI scorecards.

Cont. Data Lifecycle PoV

Lifecycle Stage	Data Science	Data Analytics	Business Intelligence (BI)
1. Data Generation & Collection	Defines requirements for raw, multi-modal, and unorganized streaming data (text, images, logs, APIs, sensors).	Relies on existing operational databases, transactional logs, and structured third-party datasets.	Relies on heavily structured, pre-filtered data from production databases and enterprise applications.
2. Data Storage & Ingestion	Works with Data Lakes, NoSQL, and Big Data frameworks (Hadoop, Spark, S3) handling high-volume, unstructured data.	Works with Data Warehouses, relational databases (SQL), and curated flat files (CSVs, Excel).	Works exclusively with structured Data Warehouses, Data Marts, and OLAP cubes built via ETL pipelines.
3. Data Processing & Cleaning	Conducts complex feature engineering, tokenization, missing-data imputation, and advanced programmatic wrangling.	Focuses on data cleaning, filtering, joining tables, transforming variables, and handling outliers for specific questions.	Consumes clean, pre-aggregated, and standardized data; performs light aggregation and metric formatting.
4. Data Analysis & Modeling	Builds, trains, and tunes predictive statistical models, Machine Learning algorithms, and Deep Learning architectures.	Applies diagnostic statistical methods, hypothesis testing, correlation analysis, and trend identification.	Executes descriptive aggregations, KPI calculations, grouping, slicing/dicing, and historical tracking.
5. Visualization & Delivery	Ships production-ready ML pipelines, automated decision APIs, model artifacts, and custom interactive notebooks.	Delivers ad-hoc analytical reports, diagnostic presentations, data stories, and actionable business recommendations.	Publishes static or real-time executive dashboards, automated daily/weekly reports, and KPI scorecards.
6. Action & Decision Implementation	Enables **automated / continuous decision-making** (e.g., dynamic pricing, recommendation engines, fraud blocking).	Guides **tactical & strategic decisions** via human interpretation of analytical findings and recommendations.	Supports **operational monitoring & executive review** to verify business health against established goals.

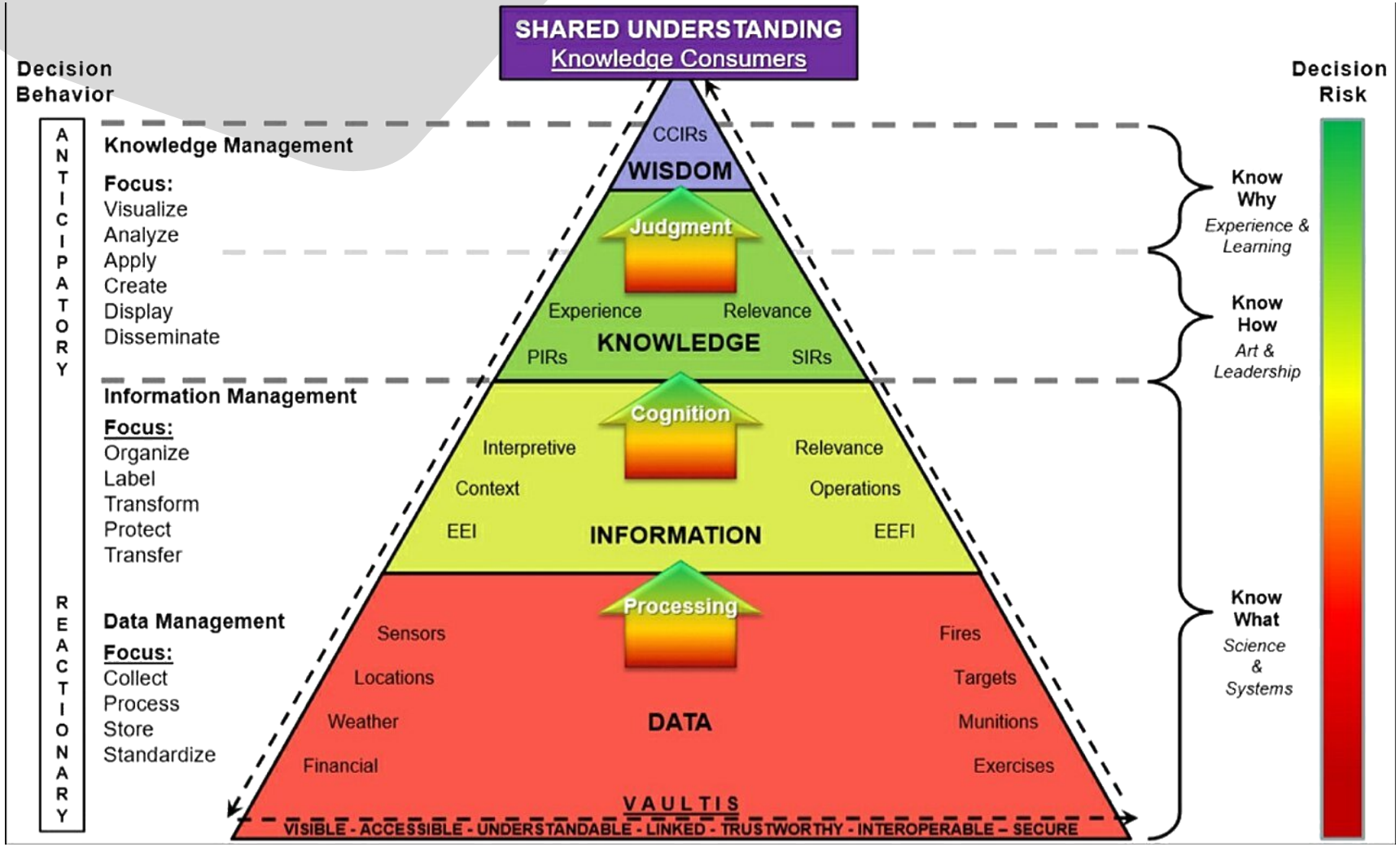
Utilitarian viewpoint



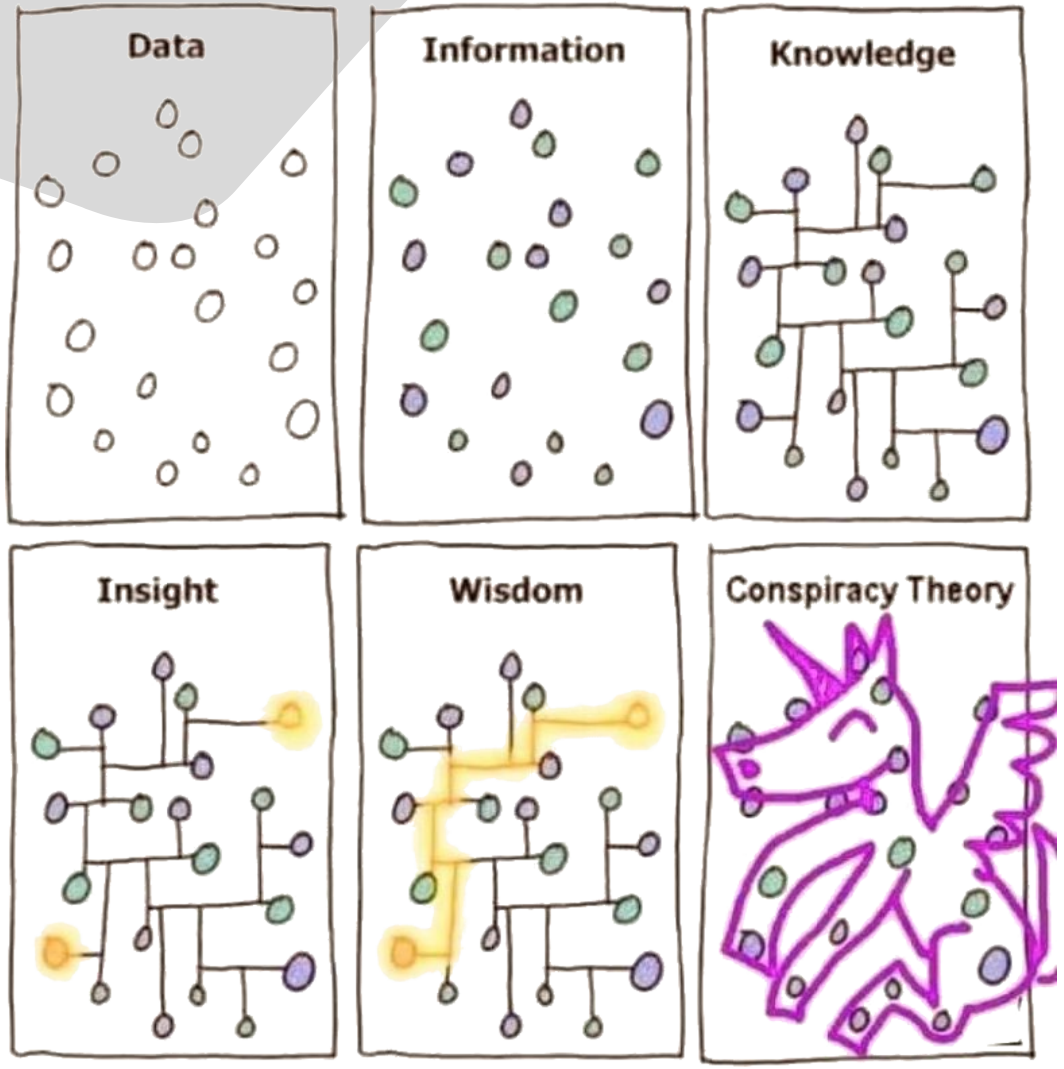
Definition, again. More *objectively*

- Data: something recorded/written down
- Science: organizes knowledge through observation, experimentation, and logical reasoning
- the only difference between messing around and science is writing it down. Adam Savage

Data evolution (US DoD)



Data evolution: some guy on linkedin



What to do

- Data collection/scrapping
- Data cleaning/wrangling
- Exploratory Data Analysis
- Feature *Engineering* and Pattern Recognition
- Model/System Building, Deployment, and Testing
- Communication and Presentation

Where to start?

